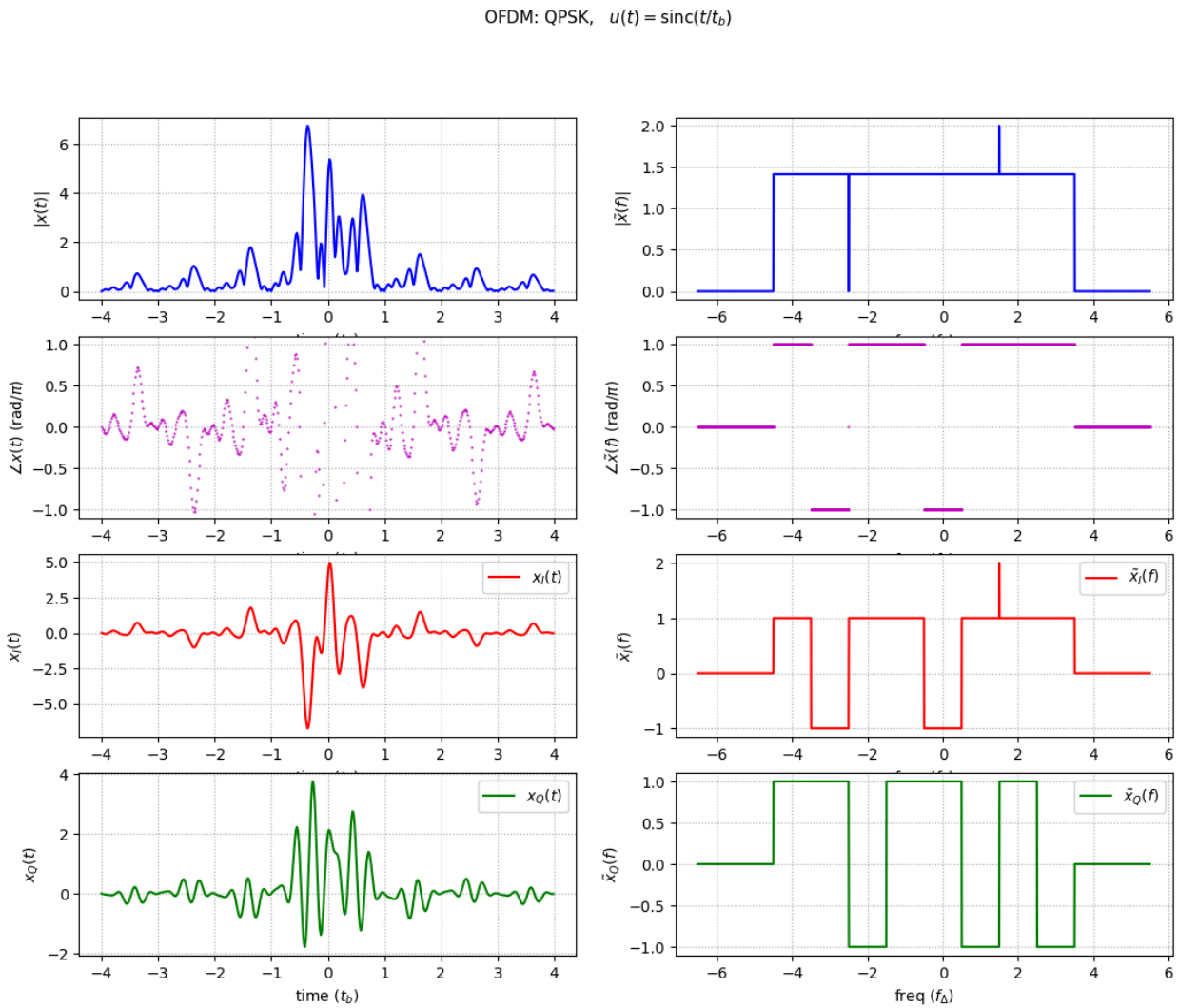
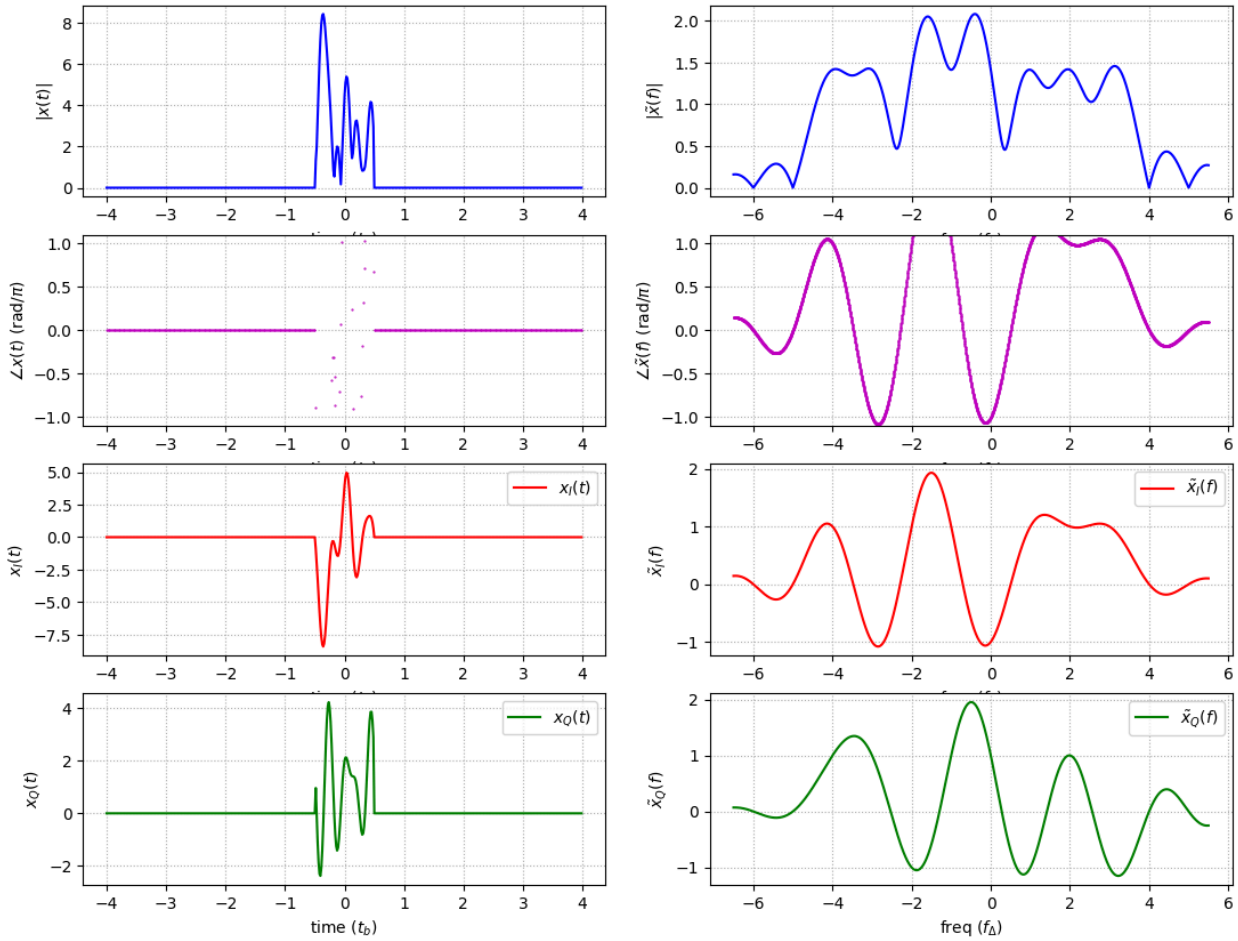
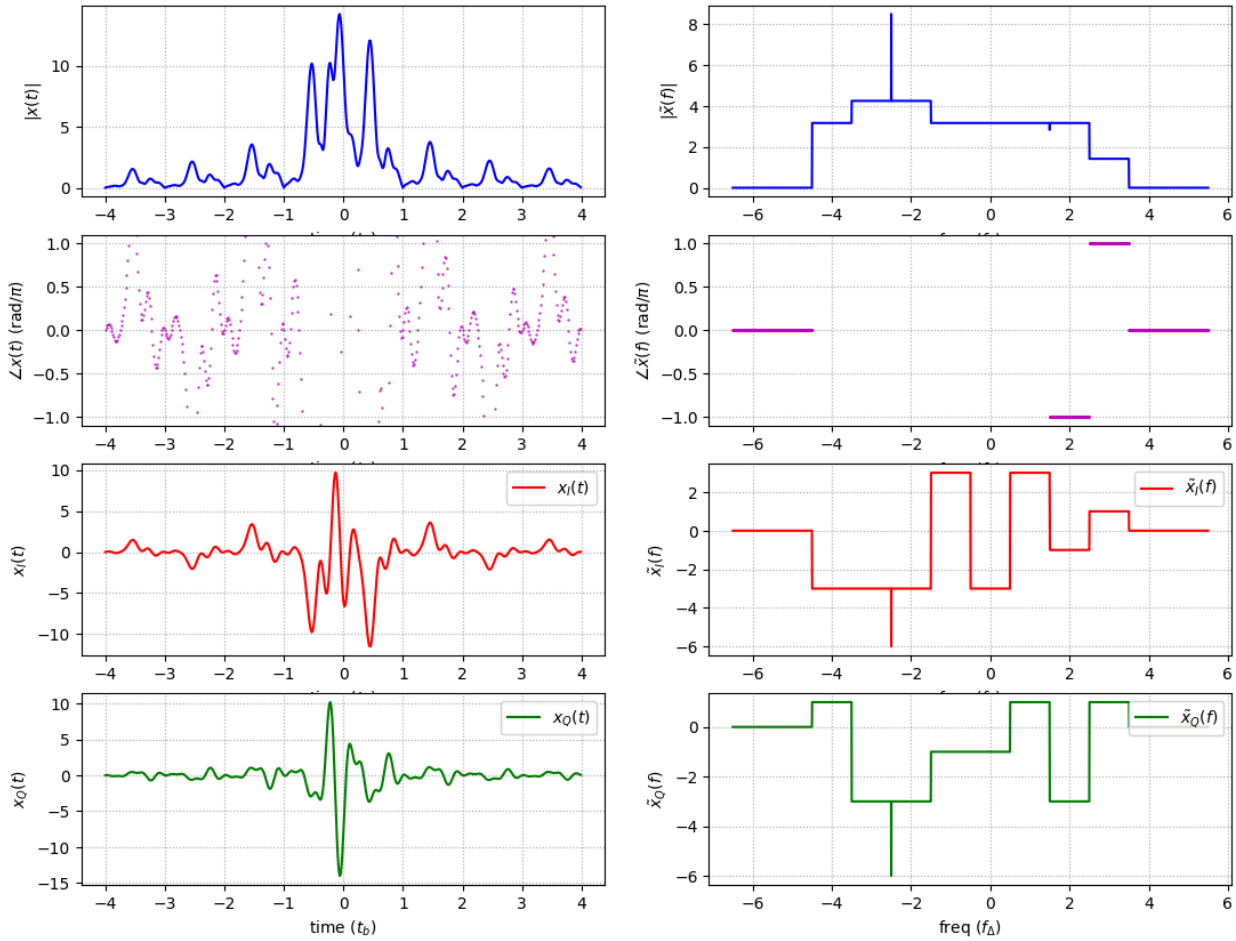
Problem 2.2.A.6



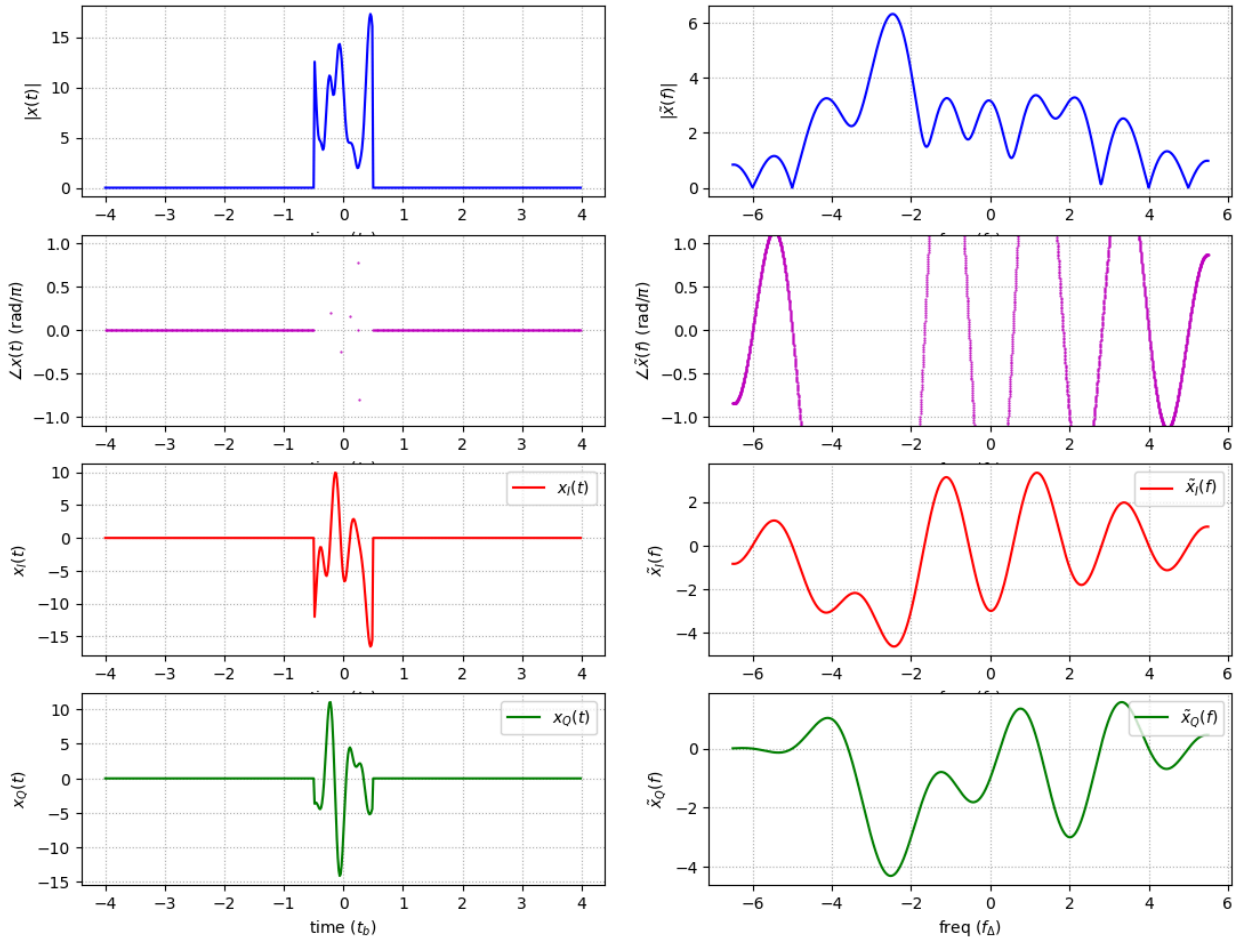
OFDM: QPSK, $u(t) = \mathrm{rect}(t/t_b)$ (square pulse)



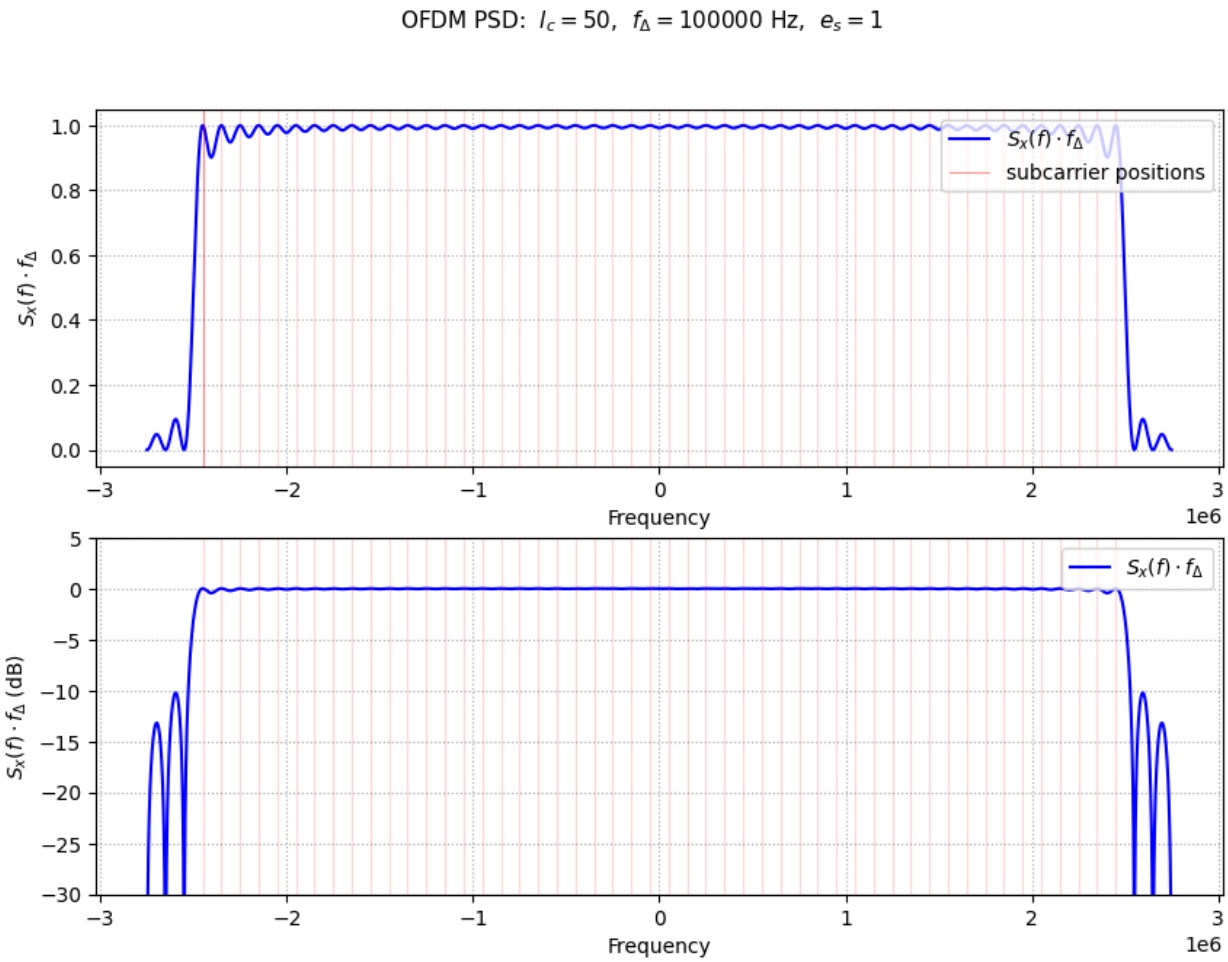
OFDM: 16-QAM, $u(t) = \mathrm{sinc}(t/t_b)$



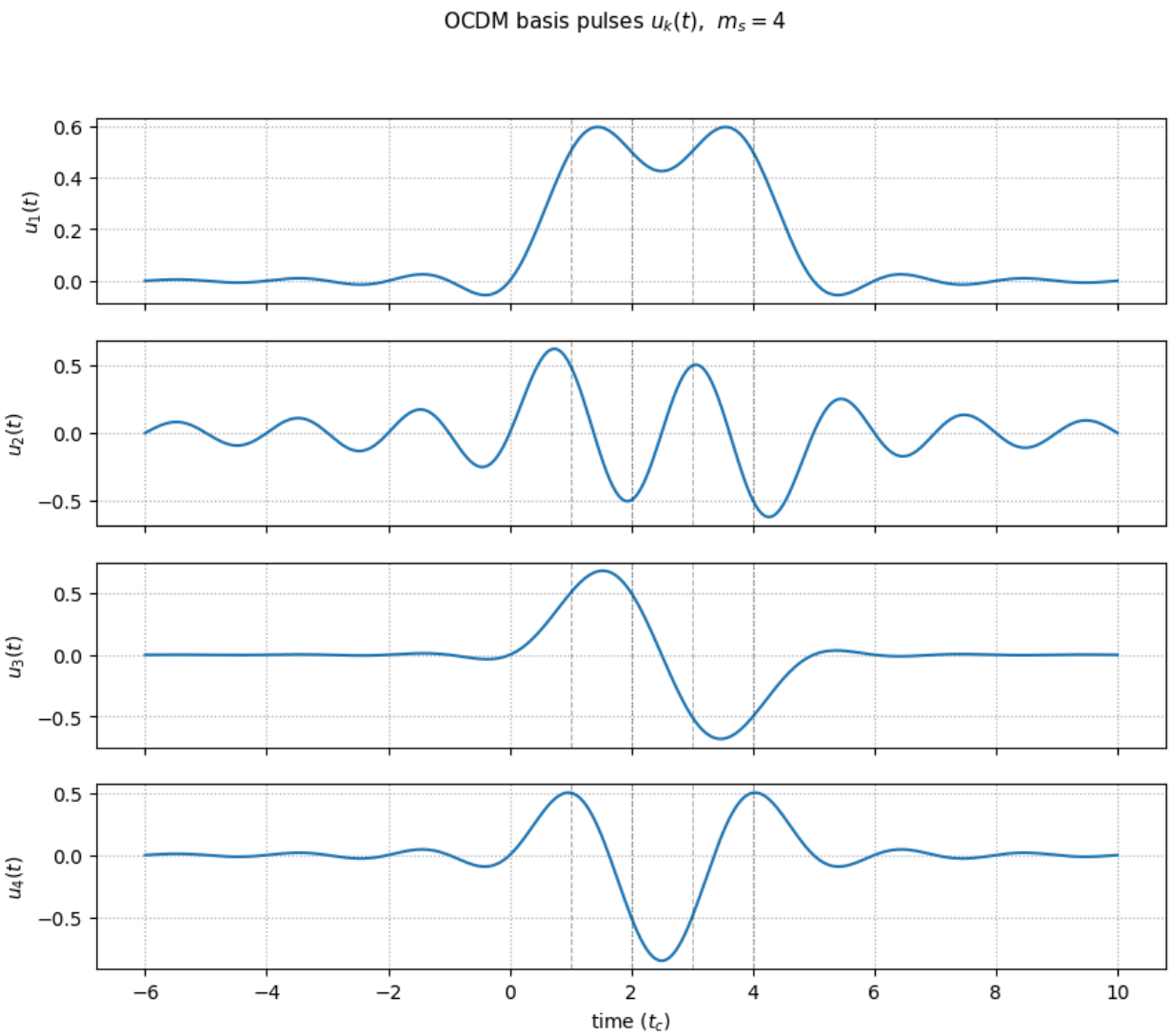
OFDM: 16-QAM, $u(t) = \mathrm{rect}(t/t_b)$ (square pulse)

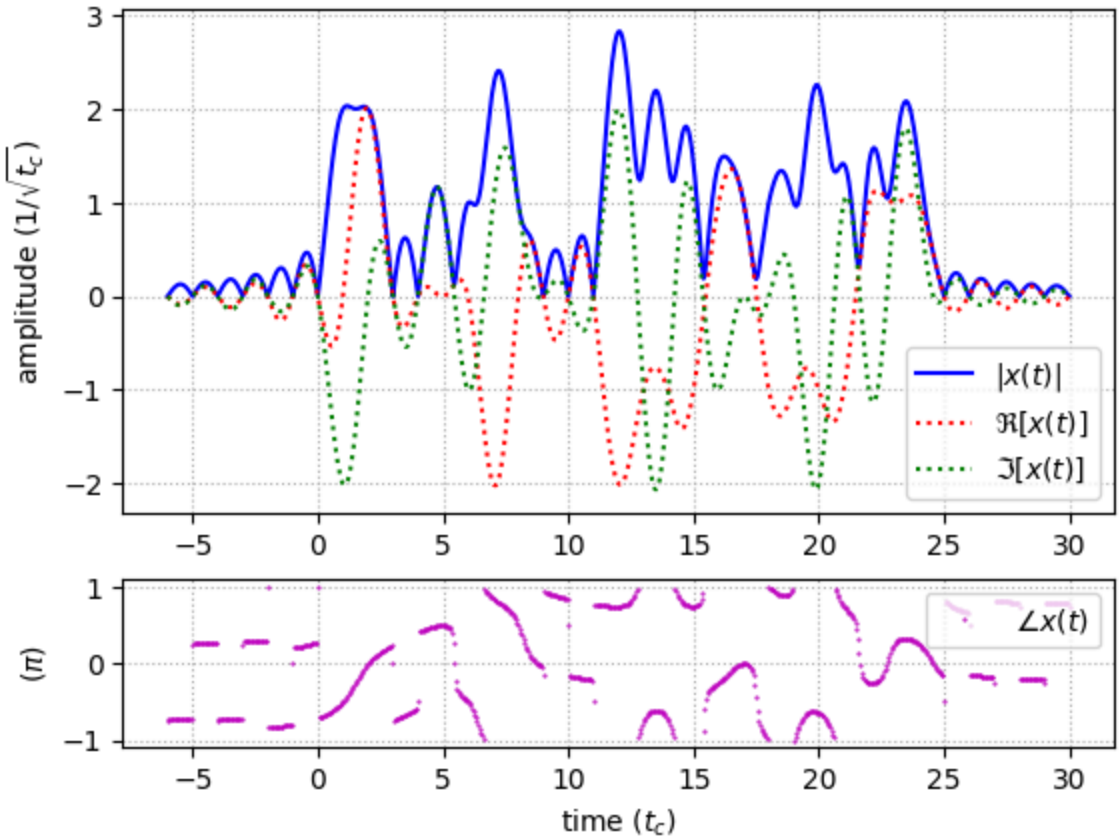


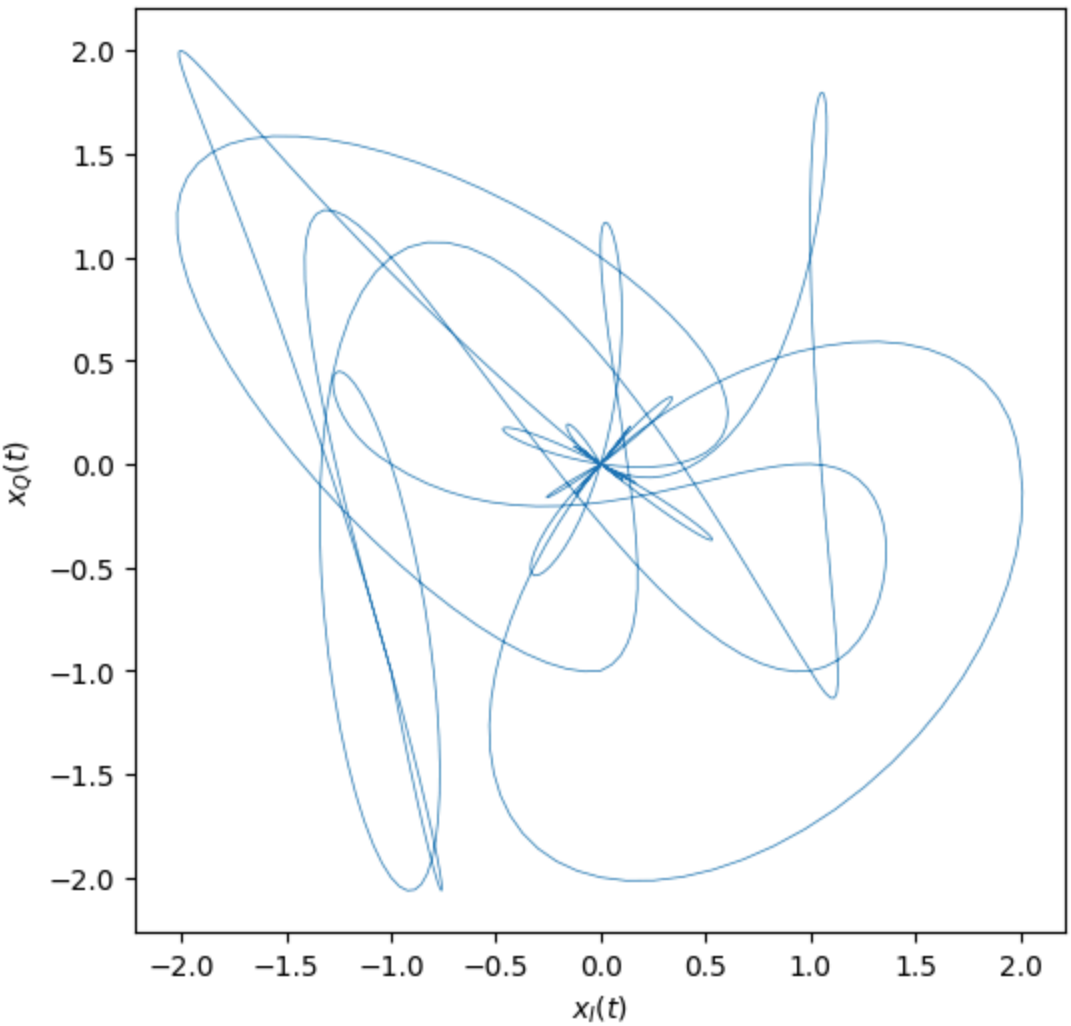
Problem 2.2.A.7

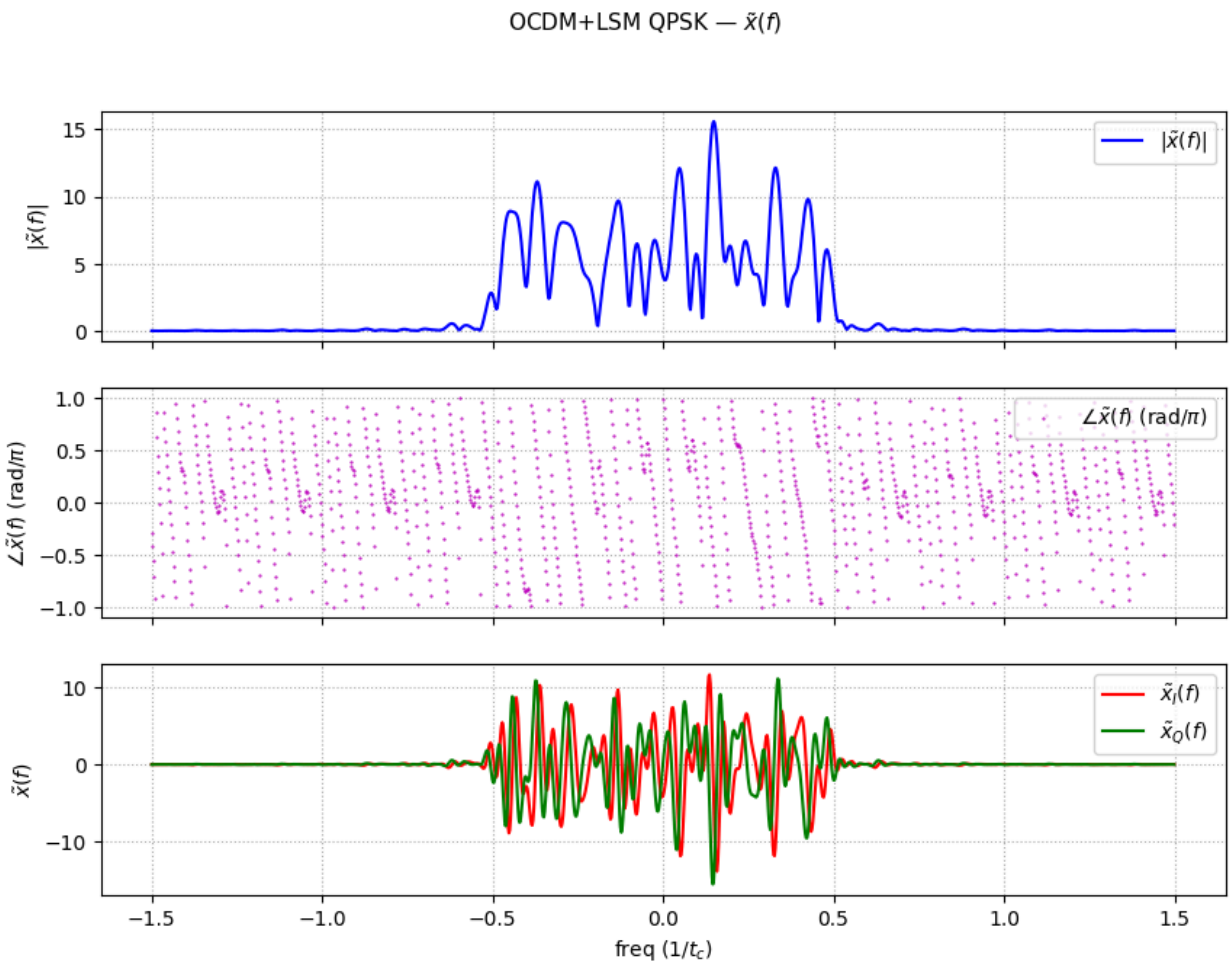


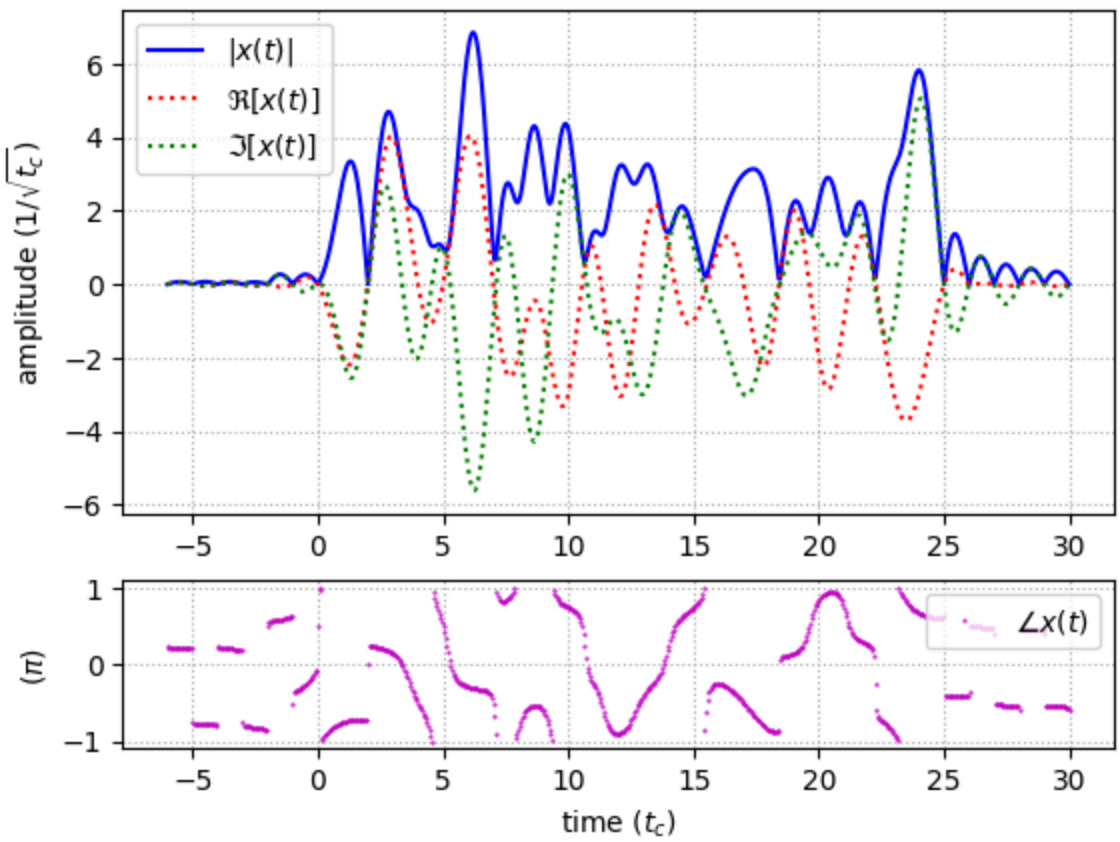
Problem 2.2.A.8

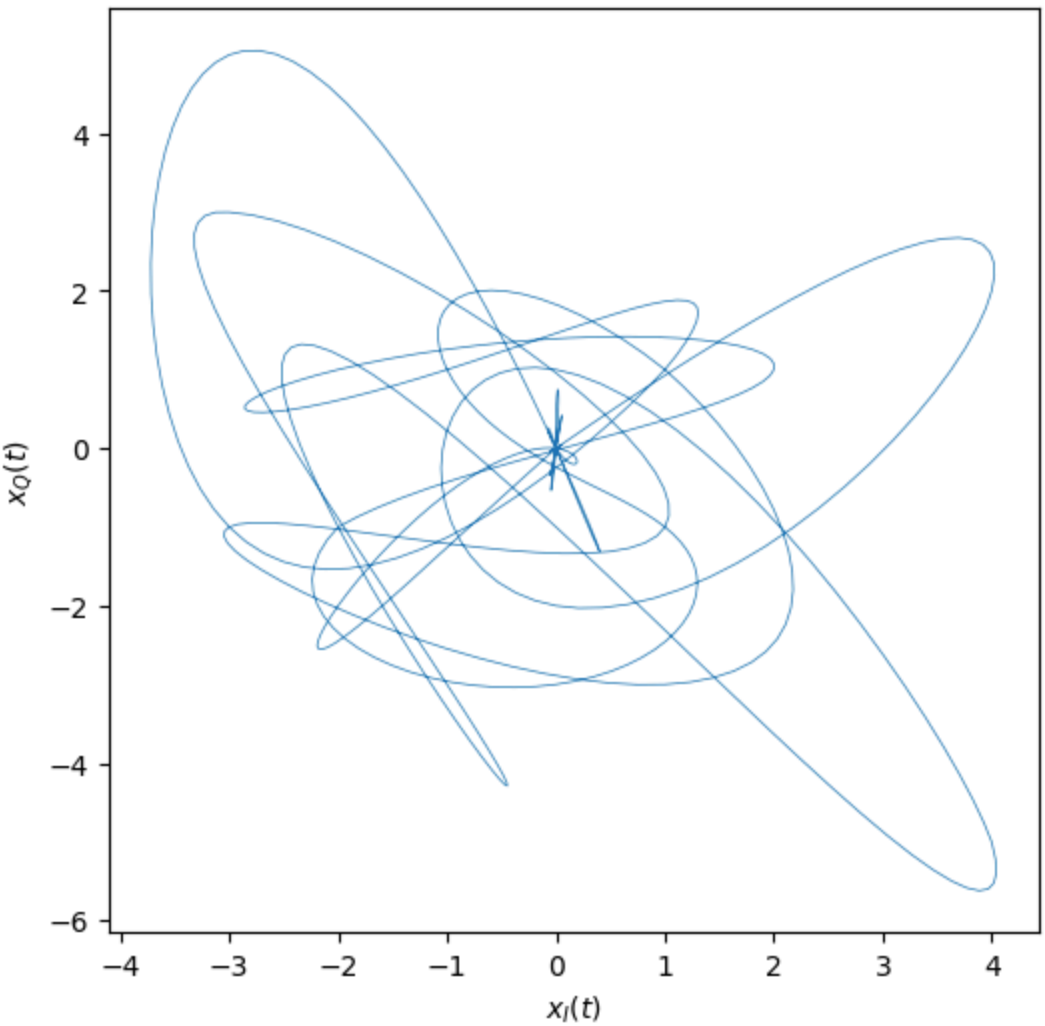

OCDM basis pulses $u_k(t)$, $m_s = 4$
$u_1(t)$
$u_2(t)$
$u_3(t)$
$u_4(t)$
0.6
0.4
0.2
0.0
0.5
0.0
−0.5
0.5
0.0
−0.5
0.5
0.0
−0.5
−6
−4
−2
0
2
4
6
8
10
time (t_c)

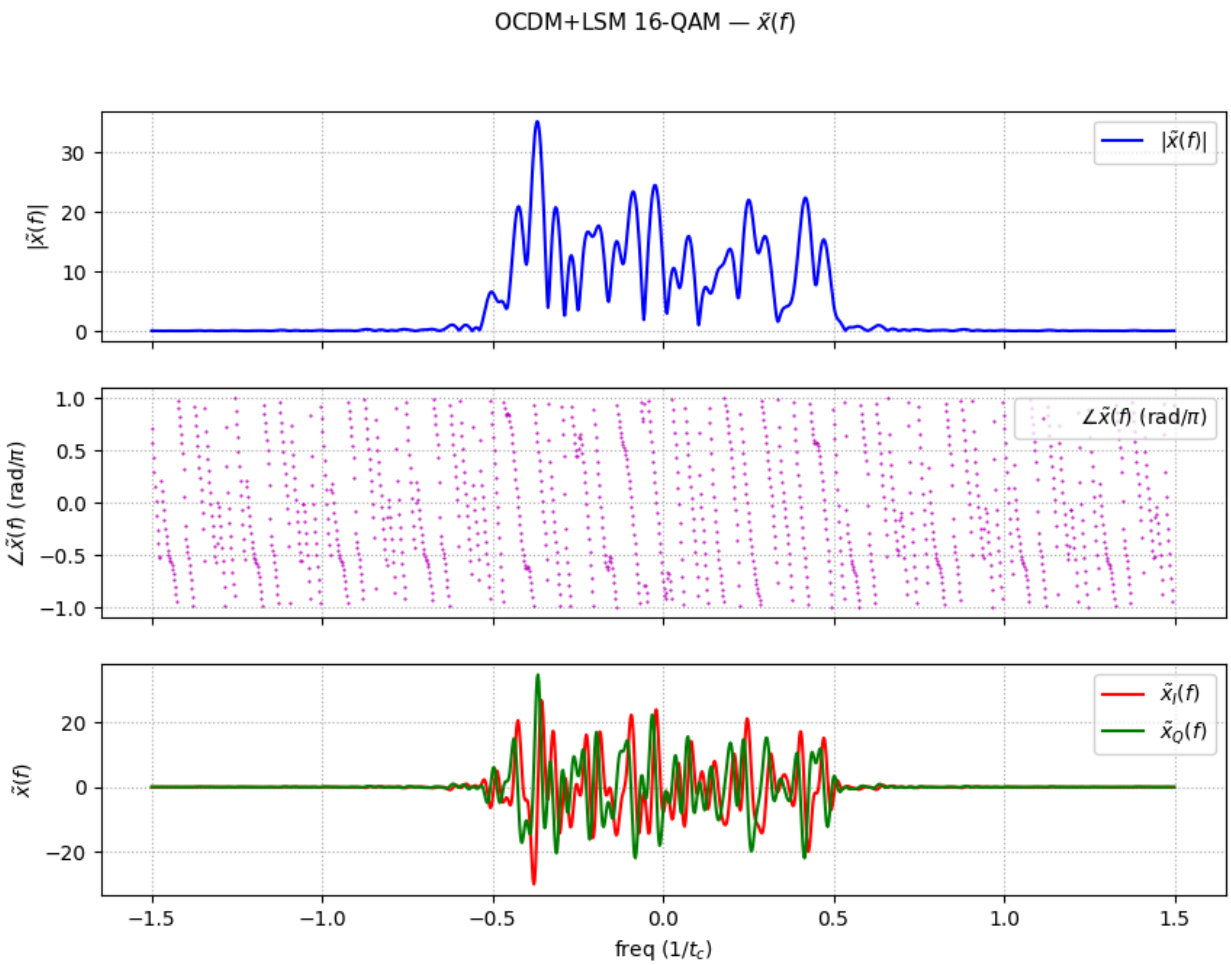

amplitude ($1/\sqrt{t_c}$)
$|x(t)|$
$\Re[x(t)]$
$\Im[x(t)]$
(π)
$\angle x(t)$
time (t_c)


2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
$x_Q(t)$
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
$x_I(t)$


OCDM+LSM QPSK — $\tilde{x}(f)$
$|\tilde{x}(f)|$
$\angle\tilde{x}(f)$ (rad/π)
$\tilde{x}_I(f)$
$\tilde{x}_Q(f)$
$\tilde{x}(f)$
freq (1/t_c)


$|x(t)|$
$\Re[x(t)]$
$\Im[x(t)]$
amplitude $(1/\sqrt{t_c})$
$\angle x(t)$
(π)
time (t_c)


4
2
0
−2
−4
−6
−4
−3
−2
−1
0
1
2
3
4
$x_Q(t)$
$x_I(t)$


OCDM+LSM 16-QAM — $\tilde{x}(f)$
$|\tilde{x}(f)|$
$|\tilde{x}(f)|$
$\angle\tilde{x}(f)$ (rad/π)
$\angle\tilde{x}(f)$ (rad/π)
$\tilde{x}_I(f)$
$\tilde{x}_Q(f)$
$\tilde{x}(f)$
freq (1/t_c)